Sachin Konan

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EDUCATION

Doctor of Philosophy, Princeton University — Fall 2025 - Current
Computer Science; Advisor: Zhuang Liu
Topic: *Towards Automated Evaluation and Execution of Scientific Discovery*

Bachelor of Science, Georgia Institute of Technology — 2018 - 2022
Computer Science; Threads in Theory and Artificial Intelligence
Research Advisor: Matthew Gombolay

RESEARCH EXPERIENCE

Meta - *Monetization (Student Researcher)* — May 2026 - Current
- *MOMO*: Developed a technique for ensembling models for scientific discovery and continual learning (In preparation for ICLR 2027)
  - Achieved a World Record on Erdos Minimum Overlap and Chip Placement
- Built a scalable, decentralized RL Post-Training Framework on Spot TPUs

Two Sigma - *Metadata* — Fall 2022 - Summer 2025
- *Functional Semantic Types*: Developed a framework to dedupe pipelines with entity detection (In NAACL 2024)
  - Finetuned LLMs for finance-specific detection and code generation
- *Synthetic Directed Graphs*: Theorized "Merge-Split", transformations that retain spectral characteristics of directed graphs while ensuring subgraph privacy
- *Compression*: Engineered 10x reduction in Two Sigma's Data Lineage Graph

Georgia Institute of Technology - *Research Assistant (CORE Robotics Laboratory)* — Fall 2020 - Spring 2022
- *InfoPG*: Developed an information-theoretic objective encouraging collaboration in multi-agent, collaborative RL games (In ICLR 2022)
- *ConDT:* Improved the architecture of Decision Transformer to discriminate between state-action embeddings based on return (In CoRL 2023)
- For *InfoPG*/*ConDT*, implemented multi-GPU training on the lab's cluster

Meta - *Fundamental AI Research (FAIR)* — Fall 2021
- *OWD:* Open World Detection for One-Stage Detection (On arXiv)

Intel - *Deep Learning Group* — Summer - Fall 2019
- Improved runtime of ArchBench (DNN hardware simulator) from 2 hrs to 4 mins with tensor caching and parallelization

PUBLICATIONS

3. Sachin Konan, Larry Rudolph, Scott Affens. "Automating the Generation of Functional Semantic Types with Foundational Models." In *The Nations of the Americas Chapter of the Association for Computational Linguistics (NAACL) 2024.*

2. Sachin Konan, Esmaeil Seraj, and Matthew Gombolay. "Contrastive Decision Transformers." In *International Conference on Robot Learning (CoRL) 2023.*

1. Sachin Konan, Esmaeil Seraj, and Matthew Gombolay. "Iterated reasoning with mutual information in cooperative and byzantine decentralized teaming." In *International Conference on Learning Representations (ICLR) 2022.*

PREPRINT/ARTICLES

4. Sachin Konan, Zhifei Deng, Qi Xu, Zhuang Liu. "MOMO: Mixture of Model Optimization." In review at *International Conference on Learning Representations (ICLR) 2027.*

3. Sachin Konan, Jonathan Liu, Zhuang Liu. "PaperScorer: How Predictable is Peer Review?" In review at *Conference on Neural Information Processing Systems (NeurIPS) 2026.*

2. Advait Gupta, **Sachin Konan**, Gabe Sarch, Zhuang Liu. "AgentVQA: A Unified Benchmark for Agentic Visual Understanding." In review at *International Conference on Learning Representations (ICLR) 2027*.

1. Sachin Konan, Kevin Liang, and Li Yin. "Extending one-stage detection with open-world proposals."

TEACHING

Princeton University - *Teaching Assistant (with Jialin Deng)* — Fall 2026 - Current
- Precept Instructor/Exam Czar for COS 418: Distributed Systems

Georgia Institute of Technology - *Teaching Assistant (with Constantinos Dovrolis)* — Fall 2021 - Fall 2022
- Head Grader for CS 3510: Design and Analysis of Algorithms

AWARDS

Princeton AI Lab Graduate Student Fellow — 2025

Thinking Machines Research Grant Sponsorship — 2025

Lambda AI Cloud Research Grant Sponsorship — 2025

NSF Graduate Research Fellowship (declined before deadline) — 2022

PURA Undergraduate Research Fellowship — 2021

NBA National Hackathon (Top 10 in the US) 2021

Davidson Fellow 2018

Finalist, Regeneron Science Talent Search 2018

2nd in Embedded Systems at Intel International Science and Engineering Fair 2017

COURSEWORK

Graduate-Level: COS 529: Advanced Computer Vision, COS 568: Systems and Machine Learning, COS 598D: Empirical Research Methods, COS 597D: Information theory in computer science, COS 597K: Frontiers in Deep Learning, CSCI-GA 3033: Deep Decision Making and Reinforcement Learning

Undergraduate Machine Learning: CS 4641: Machine Learning, CS 4803: Deep Learning, CS 4476: Computer Vision, CS 3630: Perception and Robotics, CS 3600: Intro To Artificial Intelligence

Undergraduate Computer Science: CS 4540: Advanced Algorithms, CS 3510: Design and Analysis of Algorithms, CS1302: Data Structures and Algorithms

Undergraduate Mathematics: MATH 3406: Advanced Lineage Algebra, MATH 4032: Combinatorial Analysis, ISYE 3770: Statistics, Math 2550: Into To Multivariable Calculus